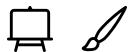


Applied Statistics: Time Series

Introduction to time series

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Definition of time series

According to you, what is a time series?

Two main ingredients: 1. Series: A “thing”, usually measurable, i.e., a number. 1. Time: This number changes over time.

Formally, a time series is a sequence of values each indexed by a **time step**:

$$X = (x_{t_1}, x_{t_2}, \dots, x_{t_N}),$$

where x_t is the value (usually a real number) of the time series at time t and $t_1, t_2, \dots$ are the time steps. Each element x_{t_i} of a time series is called a **sample**.

[Time Series](#)



Examples of time series

- Any meteorological quantity, measured over time (temperature, pressure, wind speed,...)
- Stock market prices
- Any quantity measured inside a bioreactor (temperature, pH,...)
- Any health constant measured on a patient (temperature, heart rate, blood pressure,...)

Acquisition

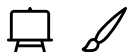
Time step

The acquisition of measurements requires the recording device to “read” the phenomenon and then “write” the recorded value. By nature, a time series cannot be continuous, because there is necessarily a gap between two consecutive values. Depending on the context, however, this gap can be made extremely small.

In practice, the duration of a time step, or sampling rate, should be made as small as needed for the purpose, but as large as possible to spare memory and computation.

Example: Recording sounds requires to sample at a frequency of approximately 20 kHz (largest audible frequency). [Time Series](#)

Actually, we need to record at twice this frequency because of aliasing.



Resampling

Sometimes, the given time step of a time series is not appropriate for our purpose.

Example: I want to study the impact of the season on temperature.
If the time step of my time series is at 15 minutes, this is too fine.

Given a fine time series, coarse graining is easy, but comes with a question. For each coarser time step, I have multiple values measured. Which one(?) should I use?

Representing

The first step in the analysis of a time series is visual inspection.

Make sure to use an appropriate plotting tool. You need to be able to rescale, zoom in, zoom out of you time series.

I recommend pyplot from matplotlib, which is a powerful plotting library.

Noise

Example: Measure the height of someone in millimeters.

Any data acquisition is imperfect. Deviations of the measurements from the “real” system’s values are called **noise**. Noise is usually added (or subtracted) from the measured values in the acquisition process.

One goal in time series analysis will be to mitigate the impact of noise.

In general, we will assume that the noise has zero mean. Therefore, the measured signal remains around the real signal.

Hands on

TODO

